

Deep learning-based automatic dairy cow ocular surface temperature detection from thermal images

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ABSTRACT

Ocular surface temperature changes can be used as indices for evaluating the health statuses of dairy cows. However, the ocular surface temperatures of dairy cows in previous studies were mostly manually obtained from infrared thermal images captured by a thermal imager. Therefore, to automatically identify health disorders in dairy cows, we propose a lightweight and high-precision detection model based on the YOLOv4 framework, named GG-YOLOv4, which is used to automatically detect ocular surface temperatures from the thermal images of dairy cows. First, the CSPDarkNet module of the backbone YOLOv4 network with a large number of parameters and high complexity is replaced by a lightweight GhostNet module. Second, depth-wise separable convolution (DepC) modules are used to replace all 3×3 standard convolutions in the neck and head of YOLOv4 to greatly reduce the numbers of model parameters and calculations. Finally, based on the correlation between the positions of the eyes and the head, a global context network (GCNet) module is introduced to obtain perfect and effective feature information to compensate for the accuracy loss induced by the light part and improve the detection accuracy of the lightweight model. To evaluate the performance of the model, the thermal infrared images produced after data expansion are used as the test dataset. The experimental results show that the mean accuracy (mAP) of the GG-YOLOv4 network model is 96.88 %, the detection speed is 40.33f/s, and the model size is 44.7 M. Compared with other target detection models, the proposed method has better comprehensive performance in terms of detection accuracy, model size and detection speed. The ocular surface temperatures of dairy cows are obtained by this detection model and compared with the ocular surface temperatures manually extracted by FLIR-Tools. The comparison results show that the average absolute temperature extraction errors in the left and right eyes are 0.051 °C and 0.042 °C, respectively, and the average relative temperature extraction errors in the left and right eyes are 0.14 % and 0.11 %, respectively. The developed method realizes the rapid and accurate positioning of the key parts of dairy cows and can be used for high-precision ocular surface temperature extraction, providing a basis for the automatic diagnosis of dairy cows' health statuses.

1. Introduction

In dairy farming, body temperature is an important parameter for measuring the health and physiological statuses of dairy cows, and it is also recognized as a key indicator for detecting animal vital signs and their heart and respiratory rates (Cuthbertson et al., 2020, Jorquera-Chavez et al., 2021). Accurate and effective cow body temperature monitoring is helpful for improving the recognition rate of cow oestrus, accurately judging the delivery periods of cows, and detecting and preventing cow fever and heat stress early, which is of great significance.

The use of a built-in digital thermometer to measure rectal temperatures is the most widely used method thus far (Burfeind et al., 2010), as it can provide the best approximation value of the core temperature of a cow. However, this method is time-consuming and requires cows to be constrained when obtaining their rectal temperatures, which may cause stress responses in cows (Naylor et al., 2012). With the development of new technologies such as subcutaneous implant launchers and vaginal recorders, these technologies can remotely and quickly record the body temperatures of cows, and exhibit certain application prospects (Koltes et al., 2018, Rose-Dye et al., 2011). However, they are equally invasive

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to a certain extent and restrain animals when used. In addition, these technologies can only obtain data after device retrieval and battery life issues need to be considered, so the data cannot be obtained in real time (Cuthbertson et al., 2019). These issues limit the application of this method in production. Therefore, a less invasive and more effective method is needed to detect the body temperatures of dairy cows.

Infrared thermography (IRT) is a non-contact surface temperature measurement method. It uses a thermal imager to measure the thermal radiation or surface temperature emitted from an object, and displays the different temperatures detected as images with each color or shadow in the image representing a different temperature (Johnson et al., 2011). The technology has been widely used to collect animal surface temperatures (Sasaki et al., 2016; Rainwater-Lovett et al., 2009). Since ocular surface temperature is highly correlated with core body temperature (George et al., 2014), locating the eye area to measure body temperature has been used as an alternative to invasive methods such as vaginal or rectal measurement approaches. In addition, in traditional studies, the surface temperatures in thermal windows are manually obtained using specific software (such as FLIR-Tools). That is, a thermal infrared image is loaded into the software, and the maximum, minimum and average temperatures in a specific area of the thermal infrared image are extracted by manually using the mouse to draw frames in the appropriate image areas. This manual method wastes considerable manpower and material resources, limiting the application of thermal infrared technology in actual production. Therefore, to detect cow eye temperatures more effectively in practice, it is necessary to develop an automatic cow eye temperature detection method that can automatically extract ocular surface temperatures from thermal infrared cow images.

With the development of artificial intelligence technologies such as deep learning, especially in image processing (object detection, semantic segmentation, etc.) (Krizhevsky et al., 2017), researchers have begun to apply neural networks to various fields. These algorithms include the faster region-based convolutional neural network (Faster R-CNN) (Ren et al., 2015), the single-shot multi-box detector (SSD) (Liu et al., 2016), You Only Look Once (YOLO) (Redmon et al., 2016), etc. Among them, as excellent one-stage detection algorithms, the YOLO series algorithms have high detection accuracies and fast detection speeds and are widely used for individual animal recognition. Jiang et al. (2019) integrated a custom filter layer by using the mean filtering algorithm and a leakage rectification function to achieve high-precision positioning for key parts such as the legs, backs, and heads of dairy cows. The precision of the algorithm was 99.18 %, the recall rate was 97.51 %, and the average accuracy was 93.73 %. Zhang et al. (2020) proposed a deep learning network, the enhanced fusion MobileNetv3 YOLOv3 (EFMYOLOv3), based on bilateral filtering-based thermal image enhancement for automatically detecting the eyes and udders of dairy cows, with an average accuracy of 96.8 %. Kang et al. (2022) used the YOLOv4 algorithm to locate cow hoof positions; the average accuracy rate reached 92.61 %, and the number of detection frames per second was 19.96. Yu et al. (2022) proposed an efficient milk cow nipple detection method based on an improved deep learning neural network called the fourth feature scale-space pyramid pooling-YOLOv4 (FS-YOLOv4) model. The average milk cow nipple detection accuracy was 98.26 %.

As mentioned above, YOLO detection algorithms have achieved good results in animal target detection, as they can detect targets in natural environments based on complex scenes. Compared with other detection models, the YOLOv4 network (Bochkovskiy et al., 2020) has been favoured by many researchers because of its superior detection performance. Therefore, based on the YOLOv4 framework, this study proposes a GG-YOLOv4 lightweight target detection model for cow eye location to meet the needs of low computing power consumption and high-precision detection in complex situations. The model selects a lightweight feature extraction network GhostNet (Han et al., 2020) as the backbone network to reduce the computational complexity and number

of parameters of the model. The model also uses depth-wise separable convolution (DepC) instead of the original 3×3 standard convolution, which further reduces the number of network parameters and memory consumption, and improves the detection efficiency of the network. In addition, to compensate for the loss of accuracy caused by model lightweight, this study introduces the global context network (GCNet) (Cao et al., 2020) at the end of the backbone network. Finally, the improved deep learning model is used to locate the eye areas of dairy cows, and the obtained position information is placed in the temperature matrix to extract their surface temperatures. The accuracy of the ocular surface temperatures of dairy cows obtained based on the GG-YOLOv4 model is verified by comparing them with the temperature information manually extracted by the FLIR-Tools software.

2. Materials and methods

2.1. Experimental data

2.1.1. Image data collection

All animal handling procedures were approved by the Laboratory Animal Welfare and Ethics Review Committee of China Agricultural University (approval number: Aw81702202-5-3). The data were collected from the Dadi Group Farm in Yanqing District, Beijing, China, in July 2021. The collection objects were mid-lactation Holstein cows that freely walked through the milking channel. As shown in Fig. 1, a 3 m-long and 1 m-wide dairy channel was connected to the entrance of the milking room. A radio frequency identification (RFID) identification device was placed at the entrance of the channel to determine the identities of the cows in the video. Two FLIR-A310 thermal imagers connected to two computers were placed on both sides of the cow channel. The camera lens was 1.8 m from the cow channel and 1.5 m from the ground to ensure that the cow's head and eyes could be photographed. The emissivity of the thermal imager was set to 0.98, which was consistent with previous studies on the eyes of dairy cows (Zaninelli et al., 2018). The HOBO U14-001 intelligent temperature and humidity

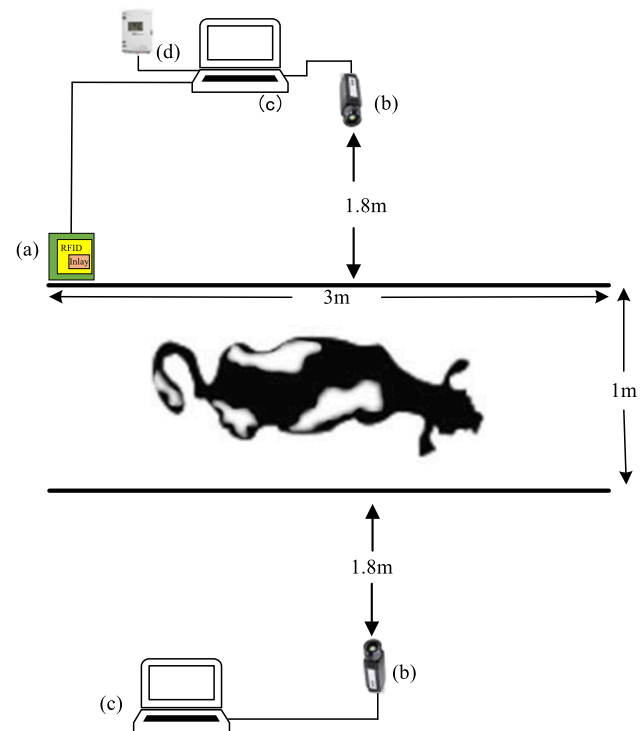


Fig. 1. Top view of the thermal infrared video acquisition platform. (a) RFID identification device. (b) FLIR-A310 thermal imager. (c) Windows 10 system computer. (d) HOBO U14-001 intelligent temperature and humidity recorder.

recorder was connected to a computer to record the ambient temperature and humidity at the place where the thermal imager was placed to correct the cow eye temperatures read by the thermal imager. When the cow was about to enter the channel, the thermal imagers were simultaneously started by the computers to collect the thermal infrared video and store the video in them.

We randomly selected 50 cows and manually extracted their left and right ocular surface temperatures to test the effectiveness of the proposed automatic detection method. The manual temperature extraction process was based on the FLIR-Tools software provided by thermal imager manufacturers. The positions of the cows' eyes were observed in the software, and frames were drawn in the eye areas. Then, the emissivity and the temperature and humidity levels in the software were adjusted, and the temperature and humidity adjustments were based on the data recorded by the temperature and humidity recorder. At this time, the software presented the maximum temperature, minimum temperature and average temperature in the region, as shown in Fig. 2. Since the maximum ocular surface temperature of dairy cows has the highest correlation with rectal temperature (Metzner et al., 2014, Chen et al., 2021), we used the maximum ocular surface temperature in the selected area provided by FLIR-Tools as the gold standard in this study to evaluate the accuracy of the automatic detection method.

2.1.2. Dataset

The specific parameters of the thermal infrared imager FLIR-A310 used in the study were shown in Table 1. The collected thermal infrared videos in the seq format were framed, and 6000 thermal infrared images with cows (in the jpg format with 320×240 pixel resolutions) were obtained. After data cleaning, some images without cow eye targets were removed, 800 images were taken from the images taken by two thermal imagers, and a total of 1600 images were obtained. Dairy cow images were marked by Labelling software, and the labelled target samples were stored in an Annotations folder in xml format to form the dairy cow thermal infrared dataset. We randomly selected 60 % of the thermal infrared images of dairy cows for model training, 20 % of the thermal infrared images for model verification, and 20 % for model testing.

2.1.3. Data enhancement

To enrich the image training set, effectively extract image features and avoid overfitting, we extended the image samples. The data

Table 1

The parameters of thermal infrared imager equipment.

Parameter name	Resolution / pixels	Frame rate / f/s	Temperature range / °C	Thermal sensitivity / °C	Accuracy / °C
Parameter range	320 × 240	10	−20 ~ 120	<0.05	±2

expansion process must meet the practical application conditions. In this study, the possible interference in target detection was considered, and three methods (motion blur, image rotation and brightness change) were mainly used for data expansion. The original 1600 thermal infrared images were expanded to 6400 images for network training and parameter optimization, ensuring the robustness of the training phase and increasing the generalization ability of the network. Considering that the cow angles in the images generated by translation, horizontal and vertical flipping were not consistent with the actual situation, this study did not use translation and flipping as image enhancement methods. Image enhancement effects are shown in Fig. 3.

2.2. Overall process flow of the proposed method

This study utilizes the following steps to achieve automatic cow eye temperature extraction: thermal infrared video framing, dataset construction, YOLOv4 model training, improved YOLOv4 model (GG-YOLOv4) construction, cow head posture division, cow eye positioning, and ocular surface temperature extraction. The flow of the proposed method is shown in Fig. 4. Studies have shown that the eyes are the hottest areas in the head of an animal (Jorquera-Chavez et al., 2020). The concept behind the proposed method is extracting this hot spot from the animal head. As shown in Fig. 4 (a), first, the collected thermal infrared videos are framed, and the dataset is established for deep learning model training. Since the original YOLOv4 algorithm contains too many detection layers, the weight is too large and the detection speed is slow, we propose a lightweight GG-YOLOv4 target detection model. Second, since the eye traits of dairy cows are not obvious, to reduce the number of incorrect eye area judgments, this study adopts the strategy of prioritizing the heads of dairy cows and repositioning the eyes of dairy cows. When a cow passes through the channel, the left and right tilt of the head posture (relative to the movement direction of the cow) change the directions of the cow eyes at different angles and

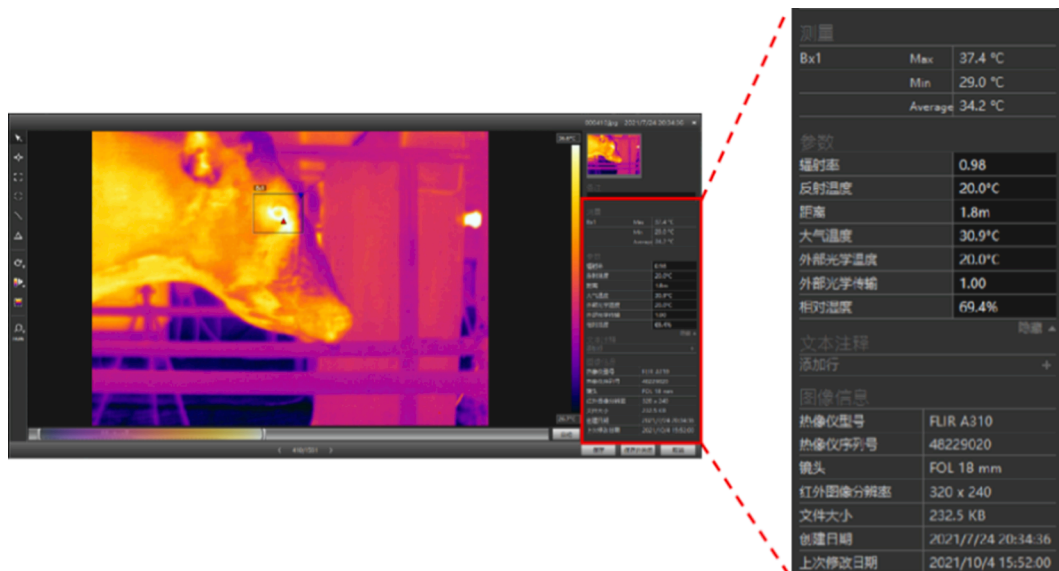


Fig. 2. Ocular surface temperatures were manually obtained using the FLIR-Tools software. According to the real experimental scene, the distance was set to 1.8 m, the atmospheric temperature was set to 30.9 °C, and the relative humidity was set to 69.4 %, and other parameters remained unchanged in FLIR-Tools software.

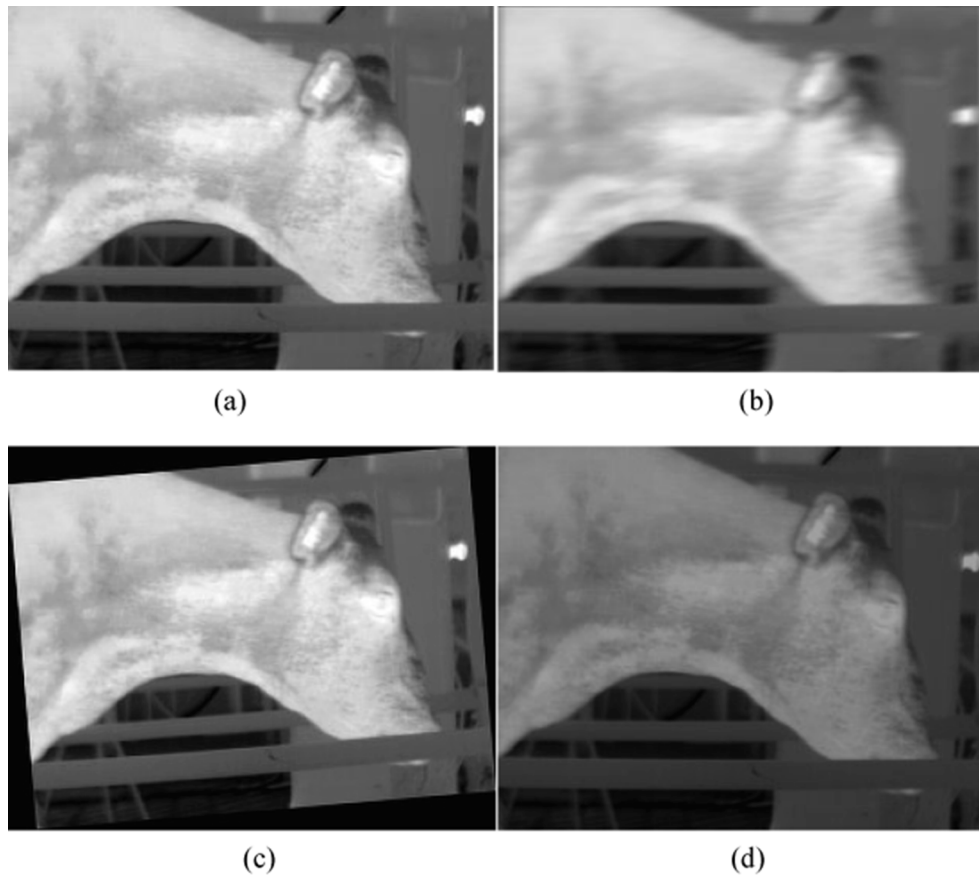


Fig. 3. Image enhancement effect diagram. (a) Original image. (b) Motion blur result. (c) Image rotation result. (d) Brightness change result.

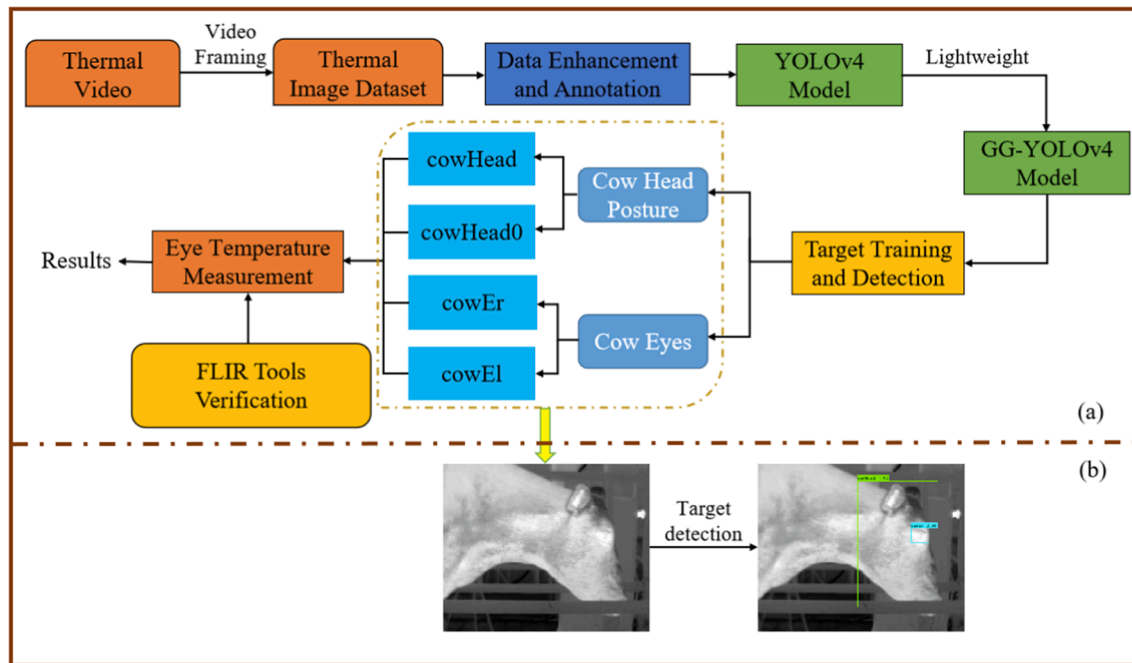


Fig. 4. System overview and purpose of the proposed method. (a) System overview. (b) Target detection.

positions, affecting the measurement of the cow ocular surface temperature (Zhang et al., 2020). Therefore, to detect ocular surface temperatures accurately and effectively, distinguishing cow head postures is necessary for future applications. In this study, the cow head postures

are divided into the upright cow head posture (cowHead) and tilted cow head posture (cowHead0). Only under the upright cow head posture (cowHead), the left ocular surface temperature (cowEl) and right ocular surface temperature (cowEr) are extracted. Finally, as shown in Fig. 4

(b), a bounding box is drawn around the eyes under the upright head posture of the cow. The coordinates of the bounding box are placed on the temperature matrix, and the temperature of the eye area is extracted as the final result, which is compared with the temperature manually extracted by the FLIR-Tools software.

2.3. Improved YOLOv4 deep learning algorithm

Different from target detection models based on candidate regions (such as Faster-RCNN), YOLO series models directly regress the locations and categories of bounding boxes in the output layer with the whole image as input, thus transforming the two-stage target detection classification problem into a single-stage regression problem. The YOLOv4 algorithm has been optimized in many aspects based on the previous YOLO detection architecture. Its main network structure includes a backbone network, a neck network and a head network. A schematic diagram of the YOLOv4 network is shown in Fig. 5.

2.3.1. Ghost module

Due to computing performance and storage space limitations and the imposed detection speed requirement, the target detection network needs to be as lightweight as possible on the basis of ensuring high precision. The Ghost module proposed by Han et al. is a lightweight convolution module derived from the feature map redundancy problem. This module does not completely abandon the standard convolution part. First, the standard convolution is used to generate a small number of feature maps, and then a Ghost feature map is generated. Through this method, the amount of calculations is reduced, and the recognition performance of the obtained features is guaranteed. It is an efficient convolution method. The standard convolution and Ghost convolution modules are shown in Fig. 6.

The Ghost module is carried out in two steps. First, a small number of feature maps are generated by a standard convolution with fewer calculations. Then, a new similar feature map is generated by further using fewer feature maps through a linear operation. Finally, the information in the two sets of feature maps is combined as the total feature information, and the generated feature map can be comparable to the feature map obtained by standard convolution (Guo et al., 2022). If the input feature map size is $H \times W \times c$, the output feature map size is $H' \times W' \times n$, and the convolution kernel size is $k \times k$, where H and W are the length

and width of the input data, respectively, c is the number of input channels, H' and W' are the length and width of the output data, respectively, and n is the number of output channels. When the input data are subjected to the standard convolution, the number of parameters is $H' \times W' \times n \times k \times k \times c$. When the input data pass through the Ghost convolution module, the number of parameters is $H' \times W' \times \frac{n}{s} \times k \times k \times c + s - 1 \times \frac{n}{s} \times H' \times W' \times k \times k$. Finally, the compression ratio between the standard convolution module and Ghost module is $\frac{s \times c}{c + s - 1} \approx s$, which greatly reduces the amount of required calculations.

2.3.2. GhostNet: Lightweight backbone feature extraction network

The improved YOLOv4 backbone uses GhostNet. Fig. 7 shows the overall architecture of GhostNet, which is mainly composed of a Ghost bottleneck stack. The Ghost bottleneck is composed of Ghost convolution layers, similar to ResNet structure, which integrates multiple convolution layers and shortcut operations. Ghost bottleneck 1 is composed of two stacked Ghost modules for the case where $\text{stripe} = 1$. The first Ghost module is used as an extension layer to increase the number of feature channels. The second Ghost module is used to reduce the number of channels, and then the inputs and outputs of these two Ghost modules are connected through a shortcut. Notably, the second Ghost module does not use the rectified linear unit (ReLU) activation function. Ghost bottleneck 2 is provided for the case where $\text{stride} = 2$ and is different from Ghost bottleneck 1. The shortcut path of Ghost bottleneck 2 is realized by a depth-wise convolution of the lower sampling layer, and $\text{stride} = 2$. Considering the accuracy of network detection, GhostNet uses a variety of combinations of Ghost bottleneck 1 and Ghost bottleneck 2 with different numbers of channels, and the accuracy level exceeds that of MobileNetV3. In addition, the computational cost and memory consumption of the model are also smaller.

2.3.3. DepC module

DepC is used to replace the standard convolution with a kernel of size 3×3 . The numbers of parameters and calculations in the standard convolution are large, while DepC can divide the standard convolution into depth-wise convolution and pointwise convolution. Different from standard convolution, depth-wise convolution splits the convolution kernel into a single channel and uses a single convolution kernel to convolute each input channel to extract features. A convolution kernel is responsible for a channel, thus obtaining an output feature map that is

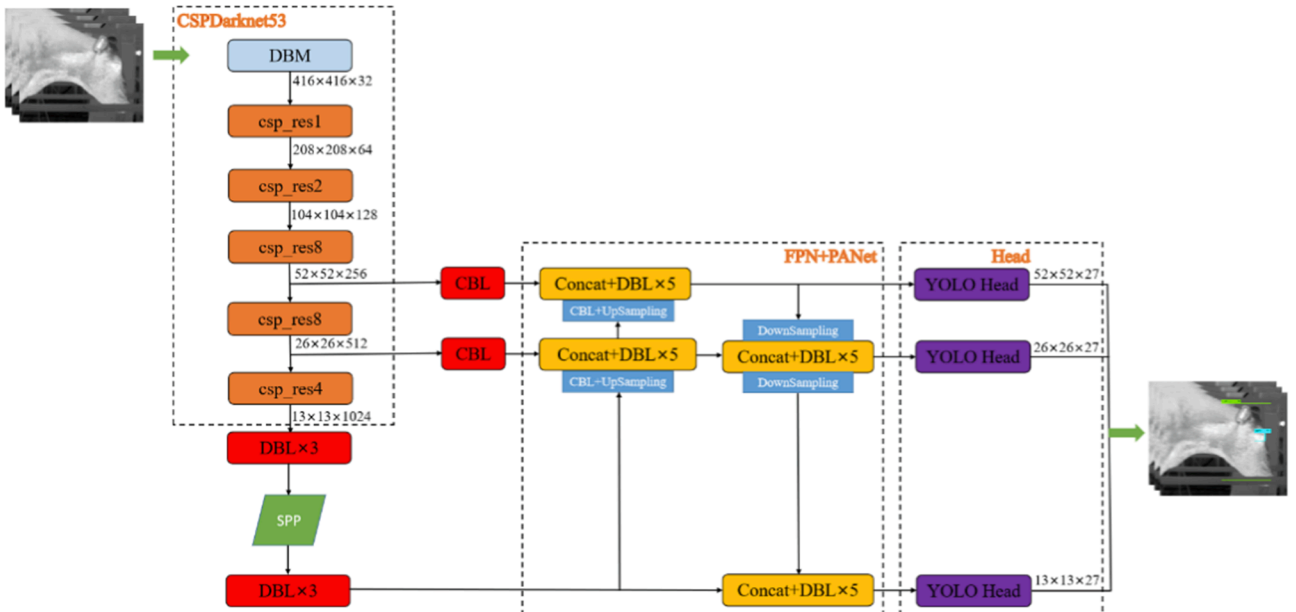


Fig. 5. YOLOv4 network structure diagram.

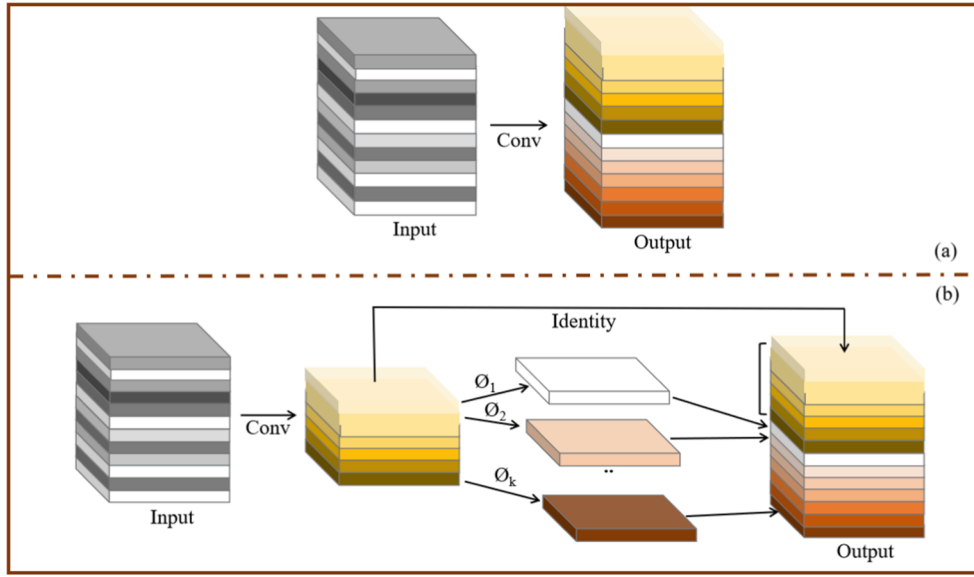


Fig. 6. Comparison between the standard convolution and Ghost convolution modules. (a) Standard convolution module. (b) Ghost convolution module. $\emptyset$ denotes a low-cost linear transformation procedure.

consistent with the number of channels in the input feature map. Pointwise convolution is actually a 1×1 convolution, which is used to linearly fuse the multiple outputs of the depth-wise convolution to realize information flow between channels. This structure can greatly reduce the numbers of calculations and parameters of the network without incurring a loss of accuracy and improve the calculation efficiency by breaking the interconnection between the output channel and the convolution kernel (Zhao et al., 2022). The DepC process is shown in Fig. 8.

2.3.4. Attentional mechanisms in GCNet

GCNet is an attention mechanism released by Tsinghua University in 2019. It combines the advantages of non-local networks (Wang et al., 2018) and the squeeze-and-excitation network (SENet) (Hu et al., 2018) and adds innovation on this basis. It combines two operations, local image feature extraction and output channel feature layer information fusion, to form a new attention mechanism. It not only has the ability to perform non-local global context modelling but can also be as lightweight as SENet, which effectively solves the problem regarding the use of a large number of network parameters that are difficult to optimize and the limitation concerning the information extraction distance. The structure of GCNet is shown in Fig. 9.

As an attention module, GCNet aggregates the location information of the global context to search for each target location in the given image. Therefore, we consider incorporating this mechanism into the YOLOv4 target detection framework. The locations of the cow head and eyes are spatially dependent, but the GhostNet feature extraction module leads to the lack of feature information for the detected object and the scattering of most information in each location of the image. After inserting the GCNet module, we capture the spatial location dependencies of the two features through non-local operations, search for the locations of each feature, and then assemble this feature information into the output feature layer. As shown in Fig. 10, the feature information enters the feature fusion module after the features of three scales are extracted by the GCNet module.

2.3.5. GG-YOLOv4 network model

This paper presents an improved target detection model called GG-YOLOv4. An image is input into the improved YOLOv4 backbone network with the lightweight GhostNet to extract the feature information of the target. Then, DepC is used to replace all 3×3 standard

convolutions in the whole model, which greatly reduces the amount of model calculations and enables the model detection more real-time. Finally, we detect the spatial location dependencies of the eyes and heads of the cows in the target and introduce the GCNet attention mechanism. When calculating the output of each pixel position, we not only calculate the neighbourhood but also correlations with all positions in the image. This approach not only ensures a small number of calculations but also effectively integrates the global information, which further improves the extraction ability of the multi-scale network for complex features. Under the premise of ensuring the integrity of the target feature information, the diversity of the model is increased. The improved model not only has a faster detection speed but also has a stronger image feature extraction ability, which improves its accuracy for the key parts of dairy cows in thermal infrared images. The improved GG-YOLOv4 network model structure is shown in Fig. 11.

2.4. Model evaluation indicators

To verify the effectiveness of the algorithm, performance evaluation indices are needed for model evaluation purposes. In this study, we use the precision (P), recall (R), F1 score, mean accuracy (mAP) and detection speed metrics as evaluation indices. The P, R and F1 score formulas are shown in Eqs. (1), (2) and (3), respectively, and the mAP is shown in Eqs. (4) and (5).

$$P = \frac{TP}{TP + FP} \quad (1)$$

$$R = \frac{TP}{TP + FN} \quad (2)$$

$$F1 \text{ score} = \frac{2 \times P \times R}{P + R} \quad (3)$$

$$AP = \int_0^1 P \times R \, dr \quad (4)$$

$$mAP = \frac{\sum_{i=1}^c AP(i)}{c} \quad (5)$$

where TP is the number of positive samples that are correctly predicted, FN is the number of negative samples that are incorrectly predicted, and FP is the number of positive samples that are incorrectly

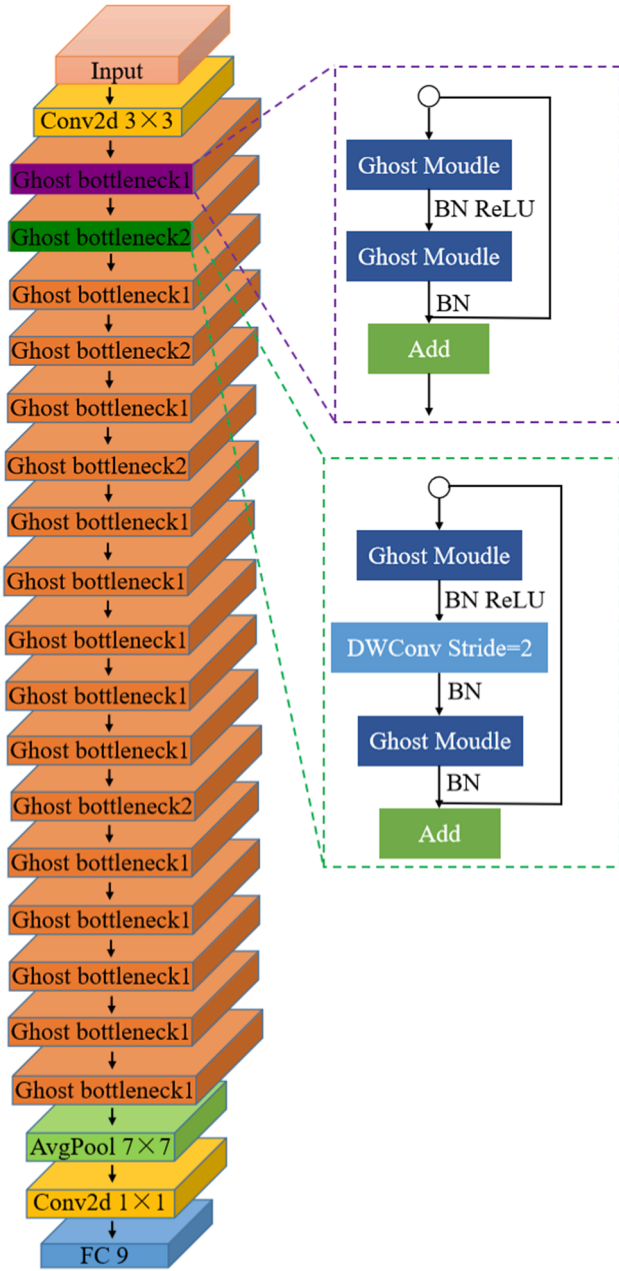


Fig. 7. Structure of GhostNet.

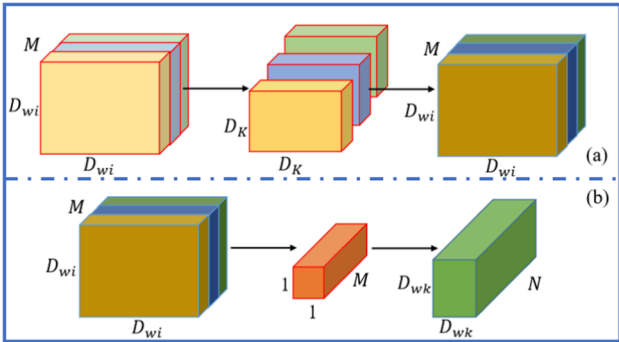


Fig. 8. DepC process diagram. (a) Depth-wise convolution. (b) Pointwise convolution.

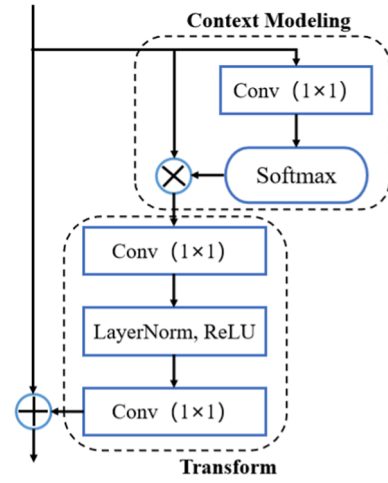


Fig. 9. GCNet module.

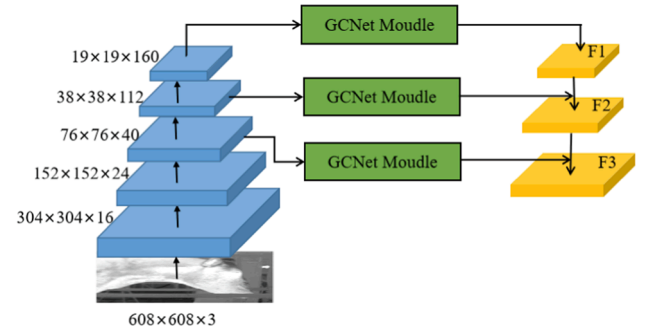


Fig. 10. GCNet module insertion location.

predicted. AP is the detection accuracy of a single category, mAP is the detection accuracy across all categories, and c is the number of detected categories.

2.5. Acquisition of cow eye temperatures

The skewness of a cow head posture affects the accuracy of cow ocular surface temperature detection. Therefore, to obtain accurate ocular surface temperatures, this paper proposes an ocular surface temperature detection strategy. When the cow passes through the channel, only the temperature data in the left and right eye image detection boxes belonging to the 'cowHead' category are extracted. In addition, a cow can produce multiple thermal infrared images through the thermal infrared lens. In this study, the average value of the maximum temperature in the thermal infrared images of the left and right eyes is used as the representative temperature of the cow's eye area to reduce the influence of the angle and distance between the cow and the thermal imager on the temperature extraction. The representative temperatures automatically extracted by deep learning and manually extracted by FLIR-Tools can be calculated by Eqs. (6) and (7) and Eqs. (8) and (9), respectively.

$$T_L = \frac{\sum_{L=1}^{N_L} \text{Max}(T_l(x, y))}{N_L} \quad (6)$$

$$T_R = \frac{\sum_{R=1}^{N_R} \text{Max}(T_r(x, y))}{N_R} \quad (7)$$

$$T_{LM} = \frac{\sum_{LM=1}^{N_{LM}} \text{Max}(T_{lm}(x, y))}{N_{LM}} \quad (8)$$

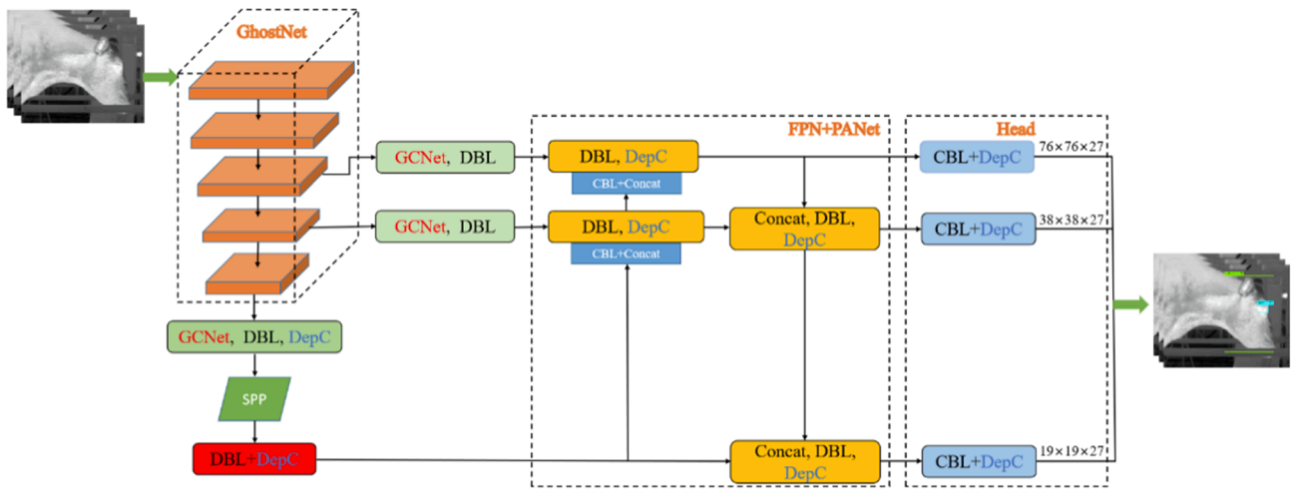


Fig. 11. Network structure diagram of GG-YOLOv4.

$$T_{RM} = \frac{\sum_{RM=1}^{N_{RM}} \text{Max}(T_{rm}(x, y))}{N_{RM}} \quad (9)$$

where T_L and T_R are the representative temperatures of the left and right eyes of dairy cows automatically extracted based on the GG-YOLOv4 model; T_{LM} and T_{RM} are the representative temperatures of the left and right eyes of dairy cows extracted manually; $\sum$ is the summation formula; N_L and N_R are the numbers of the different temperature values in left and right eyes extracted automatically under the 'cowHead' label, respectively; N_{LM} and N_{RM} are the numbers of the different left and right ocular surface temperature values obtained manually; $\text{Max}()$ is the maximum temperature function of the detection frame (manual calibration frame) in each frame image; $T_l(x, y)$ and $T_r(x, y)$ are the temperature values corresponding to each pixel in the left and right eye automatic detection boxes; $T_{lm}(x, y)$ and $T_{rm}(x, y)$ are the temperature values corresponding to each pixel in the left and right eye manual calibration boxes.

3. Results and discussion

3.1. Experimental setup

In this study, the hardware platform configuration for experimental training and testing is shown in Table 2.

To obtain better model performance, the model is trained for 200 epochs. The maximum learning rate is 0.01, the minimum learning rate is 0.0001, and the weight deviation is 0.0005. Considering the limitation of GPU memory during training, batch_size is set to 16 during the freezing training process in the first 100 epochs and 8 during the thawing training process in the last 100 epochs. Since YOLOv4 has 5 down-sampling operations in the early stage, the sizes of the input image should be multiples of 32. In this study, the resolution of each input image is set to 608×608 . The training parameters of the other models are consistent with these.

Table 2
Experimental environment.

Configuration	Parameter
CPU	Intel(R) Xeon(R) Silver 4110
GPU	NVIDIA Quadro P5000
Development environment	Python 3.8
Operating system	Ubuntu 18.04
Operating framework	PyTorch 1.8.1
Accelerated environment	CUDA 10.1 + CUDNN 7.6.1

3.2. Effect of data enhancement on detection performance

To verify the effect of the image enhancement technology on the detection of key parts of dairy cows, image datasets produced with and without data enhancement are used to train the GG-YOLOv4 network. Table 3 shows the comparison results. Although the GG-YOLOv4 network adopts data enhancement methods such as translation and mosaic enhancement during data processing, the recall rate, mAP and F1 score of the model trained on the dataset without image enhancement are low, indicating that the model trained on the dataset without data expansion is not sufficient for the detection of key parts of dairy cows. In contrast, the dataset produced using image enhancement expands the number of basic training samples, resulting in a large improvement in the performance of the trained model, which shows that image enhancement is more conducive to the accurate detection of cow heads and eyes in this study.

3.3. Performance comparison among different backbone networks

YOLOv4 uses CSPDarkNet53 as the backbone extraction network. This design method produces a large number of redundant nodes in the network and improves the accuracy of the model by sacrificing its operating speed. To select the most suitable lightweight backbone network, we evaluate several lightweight feature extraction networks, including the MobileNetv1, MobileNetv2, MobileNetv3, Densenet121 (Huang et al., 2017) and GhostNet backbone networks. In this experiment, the P, R, mAP, and model size metrics and the number of calculations are used as evaluation indices. The experimental results are shown in Table 4. It can be seen from Table 4 that the network model based on the lightweight feature extraction network achieves good results, with an mAP above 89 %. However, with GhostNet as the backbone network, the detection accuracy is not only higher than those obtained with the MobileNet series (Howard et al., 2017) of backbone network, but its computational complexity and output model size are also smaller than those of the MobileNet series. This is because compared with the MobileNet series models, the GhostNet model can better use the correlation and redundancy between feature maps with fewer calculations to achieve better detection performance. Compared

Table 3
Effect of image enhancement on model detection performance.

GG-YOLOv4	P (%)	R (%)	mAP (%)	F1 score (%)
Dataset without image enhancement	93.04	75.04	90.62	80.25
Dataset with image enhancement	93.35	90.58	96.88	91.75

Table 4
Influences of different backbone networks on detection performance.

Model	Backbone	P (%)	R (%)	mAP (%)	Amount of calculations (GFLOPs)	Model size (M)
YOLOv4	MobileNetv1	88.02	77.1	89.73	62.1	166.6
	MobileNetv2	88.71	84.82	93.47	57	161.8
	MobileNetv3	91.58	85.58	93.43	55.81	169.4
	Densenet121	90.71	88.78	94.19	95.79	177.8
	GhostNet	90.06	87.98	94.08	54.67	157.6

with Densenet121, although GhostNet has a slight loss in accuracy, it is more lightweight in terms of calculation and model size. In summary, GhostNet meets the requirements of high precision and lightweight detection of the model. In this study, GhostNet is selected to replace the CSPDarkNet backbone network in the YOLOv4 model.

3.4. Detection performance comparison among different improved methods

To verify the effectiveness of the improved GG-YOLOv4 model, we use different improved versions of YOLOv4 for a model performance comparison. The results are shown in Table 5. The experimental results show that compared with the YOLOv4 model, the improved methods have certain levels of influence on the detection results of the original YOLOv4 model.

Table 5 shows that when the YOLOv4 target detection model uses the lightweight GhostNet as the feature extraction network, the network detection speed increases by approximately 70.53 %, but the detection performance decreases slightly. When using DepC instead of the 3×3 standard convolution, the detection speed is increased by 13.67 %, and the detection performance is also reduced. After adding the GCNet module, although the detection speed slightly decreases, the detection performance is greatly improved, and mAP is 3.23 % higher than that of YOLOv4+①+②. This shows that the GCNet module can better extract feature information and improve the ability of the model to detect objects. This also means that compared with the YOLOv4 model, the GG-YOLOv4 model proposed in this paper can achieve an improved detection speed while ensuring high accuracy. The last line in Table 4 represents the detection results of the GG-YOLOv4 model proposed by us. The detection effect of this model is shown in Fig. 12.

3.5. Performance comparison among different detection models

To verify the detection performance of the proposed model, we also train other detection models for comparative experiments. In the training process, the GG-YOLOv4 model and other detection models run under the same training environment, parameters and datasets. The results are shown in Table 6.

Table 6 shows that the mAPs and F1 scores of the Faster-RCNN, SSD, YOLOv5, YOLOv7, and YOLOv4 models are more than 89 % and 75 %, respectively, because these networks superimpose high-level features and low-level features, which improves their detection accuracy

Table 5
Comparison among the results of each improved method.

Model	P (%)	R (%)	mAP (%)	F1 score (%)	Detection speed (f/s)
YOLOv4	91.35	89.93	95.31	90.5	20.12
YOLOv4 + ①	90.06	87.98	94.08	89	34.31
YOLOv4 + ②	89.74	88.44	94.53	88.75	22.87
YOLOv4 + ① + ②	90.13	85.75	93.65	87.75	42.97
YOLOv4 + ① + ② + ③ (GG-YOLOv4)	93.35	90.58	96.88	91.75	40.33

① Ghost feature extraction backbone network. ② DepC. ③ GCNet.

for key parts of dairy cows. At the same time, this also shows that the cow eye detection method based on the deep learning network is feasible. In addition, it can be seen from Table 6 that in the original network structure, the detection performance of the YOLOv4 and YOLOv7 network models is better than other networks. This is because these networks extract more abundant feature information through deep-seated networks and achieve improved small target detection abilities by adding low-level features to high-level features. The detection performance of the GG-YOLOv4 network model is better than that of the other models, which is due to the use of an attention mechanism for enhancing the extraction of high- and low-level features and improving the learning ability and detection accuracy of the model for tilted cow heads. In terms of accuracy, the GG-YOLOv4 detection model reaches 96.88 %, which is 5.48 %, 4.57 %, 7.3 %, 2.4 % and 1.57 % higher than the accuracies of the Faster-RCNN, SSD, YOLOv5, YOLOv7 and YOLOv4, respectively. In terms of the F1 score, the GG-YOLOv4 model reaches 91.75 %, which is 16.25 %, 8 %, 15.25 %, 1.75 % and 1.25 % higher than those of the Faster-RCNN, SSD, YOLOv5, YOLOv7 and YOLOv4, respectively.

When evaluating the performance of the target detection model, not only the detection accuracy but also the model parameters and detection speed should be considered. Fig. 13 shows the comparisons concerning the number of parameters, amount of calculations, model size and detection speed based on the above target detection networks.

As shown in Fig. 13, as Faster RCNN is a two-stage detection model, the target detection process is divided into two stages, so its parameter amount and calculation amount are large, and the detection speed is the slowest among all detection models. In addition, in the one-stage detection model comparison, the proposed GG-YOLOv4 network possesses 11.04 M parameters, while the SSD, YOLOv5, YOLOv7 and YOLOv4 possess 24.01, 21.07, 37.21 and 63.95 M parameters, respectively. The numbers of parameters in these comparative networks are approximately 2.17, 1.91, 3.37 and 5.79 times that of GG-YOLOv4. The number of calculations of the GG-YOLOv4 network is 13.91 GFLOPs, while the numbers of calculations of the SSD, YOLOv5, YOLOv7 and YOLOv4 are 247.95 GFLOPs, 45.53 GFLOPs, 94.59 GFLOPs and 127.7 GFLOPs, respectively. The numbers of calculations of these comparative networks are 17.83, 3.27, 6.8 and 9.18 times that of GG-YOLOv4. The model size of the GG-YOLOv4 network is 44.7 M, and the computational complexities of the SSD, YOLOv5, YOLOv7 and YOLOv4 are 96.6 M, 84.6 M, 149.3 M and 256.3 M, respectively. The model sizes of these comparative networks are approximately 2.16, 1.89, 3.34 and 5.73 times that of GG-YOLOv4. The detection speed of the GG-YOLOv4 network is 40.33f/s, while the detection speeds of the SSD, YOLOv5, YOLOv7 and YOLOv4 are 27.13f/s, 43.45f/s, 29.85f/s and 20.12f/s, respectively. The GG-YOLOv4 network is faster than the SSD, YOLOv7 and YOLOv4 and slightly slower than YOLOv5, but this is acceptable. This is due to the use of the GhostNet module as the backbone network and the use of DepC instead of the standard convolution, which reduces the complexity of the model, greatly reduces the computational cost and reduces the detection time.

In short, the proposed GG-YOLOv4 network achieves good results in terms of the detection accuracy, number of parameters, number of calculations, model size and detection speed. Considering the lightweight nature of the model and its detection speed, the accuracy of the detection model is ensured, and its overall performance is good, as it can achieve rapid and accurate key part detection in dairy cow images. Therefore, this study uses the GG-YOLOv4 network model to detect cow heads and eyes.

3.6. Analysis of the cow ocular surface temperature detection results

To test the accuracy of the proposed method for detecting ocular surface temperatures, the true ocular surface temperature values of 50 cows measured manually are used as evaluation criteria. The true and predicted values of the left and right ocular surface temperatures of cows

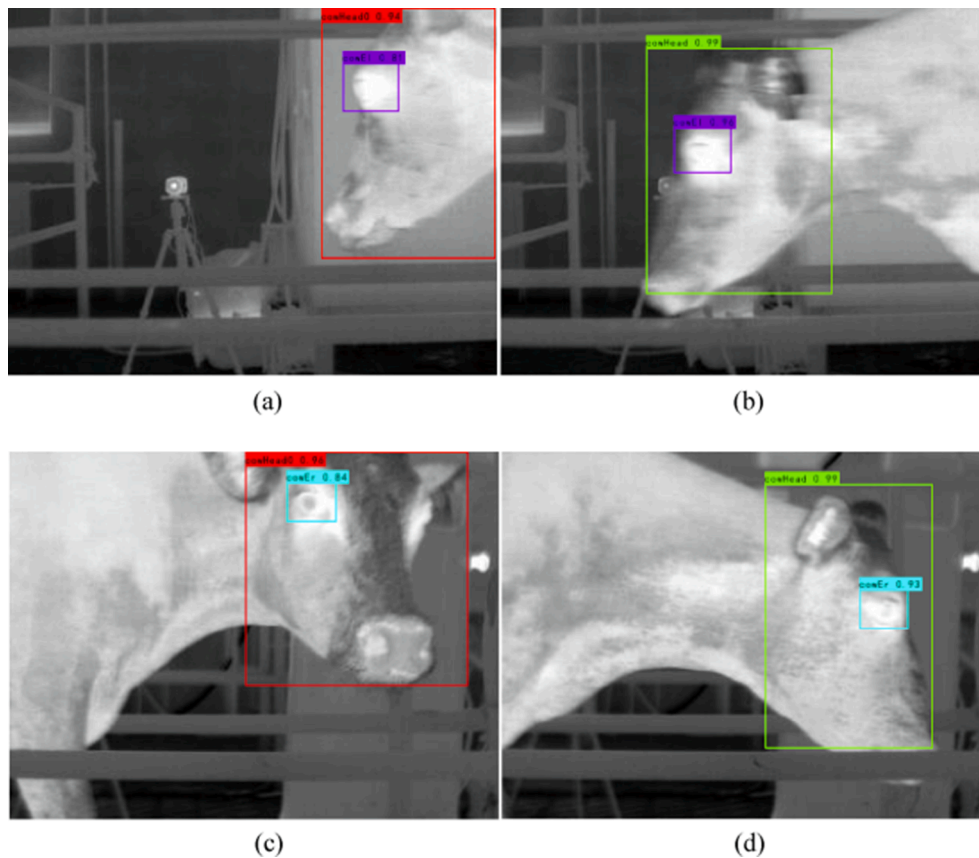


Fig. 12. Detection effects yielded for key parts of dairy cows. (a) Detection of the cow's left eye and head in a tilted posture. (b) Detection of the cow's left eye and head in an upright posture. (c) Detection of the cow's right eye and head in a tilted posture. (d) Detection of the cow's right eye and head in an upright posture.

Table 6

Accuracy comparison among different detection models.

Model	AP (%)				mAP (%)	F1 score (%)
	cowHead0	cowHead	cowEr	cowEl		
Faster-RCNN	89.28	98.22	87.35	90.76	91.40	75.5
SSD	79.75	94.12	96.70	98.66	92.31	83.75
YOLOv5	72.20	95.21	94.29	96.62	89.58	76.5
YOLOv7	85.02	96.44	98.09	98.36	94.48	90
YOLOv4	88.11	96.78	97.81	98.56	95.31	90.5
GG-YOLOv4	93.26	98.23	96.8	99.24	96.88	91.75
(Ours)						

are shown in Fig. 14. The absolute and relative errors of the true and predicted values of the left and right ocular surface temperatures of each cow are calculated. The results show that the maximum absolute error between the left ocular surface temperatures extracted by our method and the true values is 0.26°C, and the average absolute error is 0.051°C. The maximum absolute error between the right ocular surface temperatures extracted by our method and the true values is 0.22°C, and the average absolute error is 0.042°C. The maximum relative error of left ocular surface temperature detection is 0.72 %, and the average relative error is 0.14 %. The maximum relative error of right ocular surface temperature detection is 0.59 %, and the average relative error is 0.11 %. This indicates that the method developed in this paper is effective in detecting the ocular surface temperatures of dairy cows, which can replace the ocular surface temperatures in the artificially extracted infrared thermal images of dairy cows to some extent.

The method proposed in this study achieves higher accuracy than other methods. Researchers Lu et al. (2018) used a thermal imager to shoot the pig's ear root, and proposed an algorithm to automatically

extract the ear base temperature from the piglet's thermal image. By identifying the piglet's head and repositioning the ear root, the extracted ear root temperature error is within 0.4°C. He and Song (2021) proposed an automatic detection method of eye temperature of dairy cows based on thermal infrared imaging technology and a skeleton tree model. Based on the head characteristics of dairy cows and the geometric position characteristics of eyes, the eye area of dairy cows was located and the eye temperature was extracted. The average error of the detection results obtained in this study was 0.35°C, and the average relative error was 0.38 %. The main reason for this situation is that the above methods locate the region of interest of the animal through the head contour feature. Inaccurate segmentation of the animal head or head turning will cause inaccurate positioning of the region of interest, thus affecting the extraction of its temperature.

This study uses deep learning technology to realize automatic cow eye detection, and the detection accuracy achieved for cow eyes and heads is an important factor affecting the accuracy of cow ocular surface temperature extraction. Compared with other detection models (Table 6 and Fig. 13), the proposed detection model is lightweight, has high precision and induces small errors. In this study, the detection accuracies achieved for the left and right eyes of dairy cows are 99.24 % and 96.8 %, respectively. The mAP for the heads and eyes of dairy cows is 96.88 %, and the detection effect is good. In addition, the lightweight nature of the model enables it to be better applied to mobile platforms and embedded development platforms, making the model more widely applicable than other approaches.

Using thermal infrared technology to detect the ocular surface temperature of dairy cows with the model proposed in this paper provides an alternative method for evaluating animal body temperature (Macmillan et al., 2019). In this study, we divide the cow's head posture into 'cowHead' and 'cowHead0', which can not only locate the eyes of the

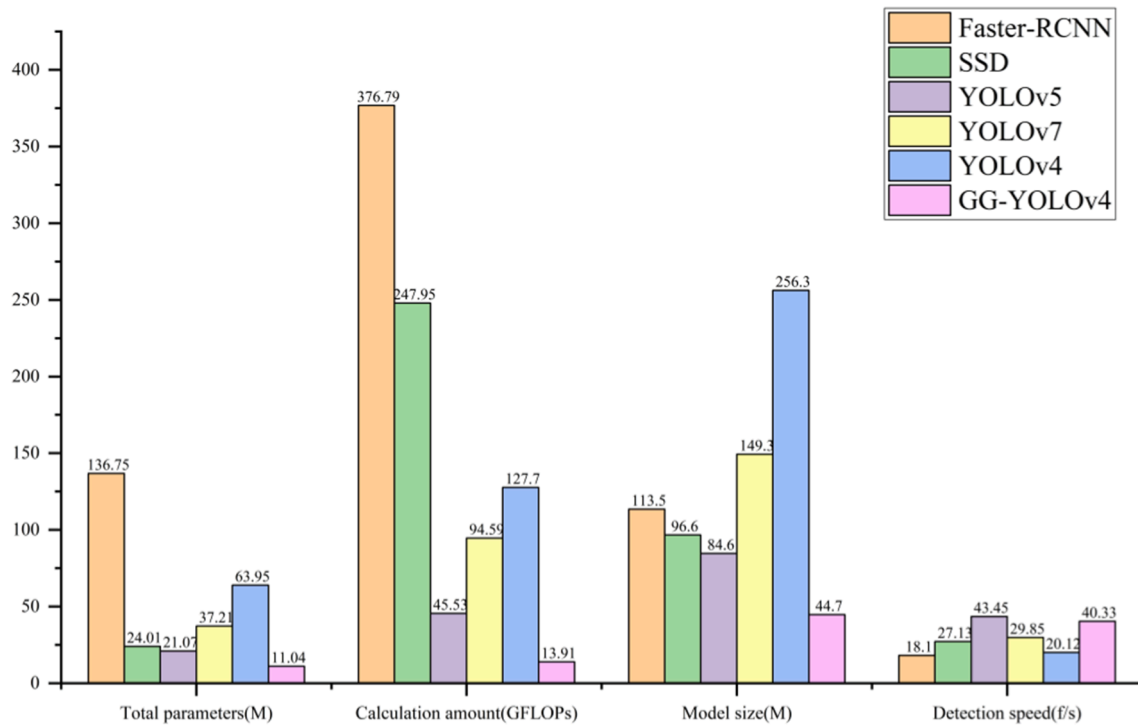


Fig. 13. Comparisons among the numbers of parameters, numbers of calculations, model sizes and detection speeds of different detection models.

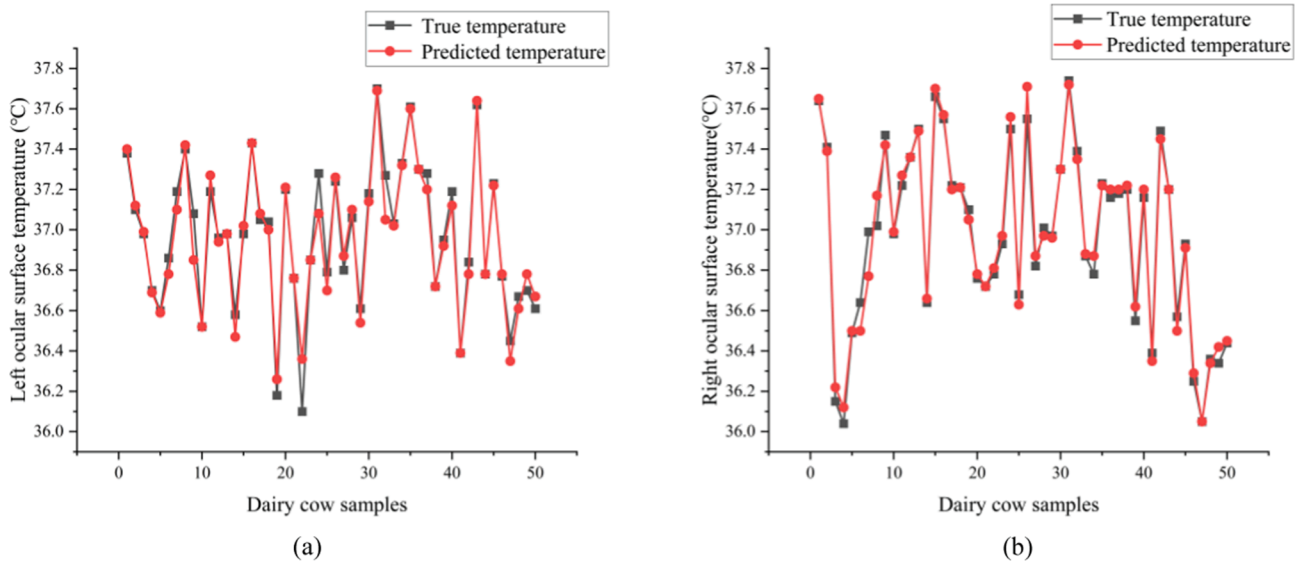


Fig. 14. The ocular surface temperature test results obtained for dairy cows. (a) Detection results concerning the left ocular surface temperatures of dairy cows. (b) Detection results concerning the right ocular surface temperatures of dairy cows.

cow head in different positions and directions (Fig. 12), but also eliminate the false ocular surface temperatures caused by incorrect head postures to ensure the accuracy of temperature extraction. In addition, the research object of this paper is the eyes and head of cows, but the proposed detection method can also be used to detect other parts of cows. New data sets are generated by calibrating other parts of the cows and detected using the GG-YOLOv4 model trained in this study. Our future work will focus on using the obtained eye temperature to monitor the health status of dairy cows, and predict cow diseases, especially inflammatory diseases and other tasks.

4. Conclusions

In this study, based on thermal infrared images of dairy cows, we propose a real-time and lightweight model called GG-YOLOv4 for detecting the key parts of dairy cows; the model is used to locate the heads and eyes of dairy cows and extract their ocular surface temperatures. The GG-YOLOv4 model solves the problems concerning the use of too many parameters and high complexity by using the GhostNet module and a DepC module. Based on the GCNet module, the model completes local and global feature extraction and achieves improved detection accuracy for key parts of dairy cows. The mAP of the model reaches 96.88 %, the F1 score reaches 91.75 %, the detection speed

reaches 40.33f/s, the number of parameters is 11.04 M, the number of calculations is 13.91 GFLOPs, and the model size reaches 44.7 M. Compared with the YOLOv4, SSD, YOLOv5 and YOLOv7 models, the GG-YOLOv4 model has better overall performance. Compared with the representative ocular surface temperatures extracted manually, the average absolute errors of the representative left and right ocular surface temperatures extracted by the proposed method are 0.051°C and 0.042°C, respectively. The average relative errors of the left and right ocular surface temperature extraction results are 0.14 % and 0.11 %, respectively. In future research work, combined with the correlation between the ocular surface temperatures and rectal temperatures of representative dairy cows in existing studies, we will use the representative ocular surface temperatures extracted by the proposed method for dairy cow health assessments.

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CRediT authorship contribution statement

Yanchao Wang: Data curation, Methodology, Software, Validation, Visualization, Writing – original draft, Formal analysis. **Xi Kang:** Investigation, Software, Validation. **Mengyuan Chu:** Investigation, Validation. **Gang Liu:** Supervision, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.compag.2022.107429>.

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